System walkthrough

AC Literature Attribution System

How the system turns broad paper search results into an AC knowledge base, then lets people query, explore, and reason over that knowledge.

Automated pipeline

ingest → classify → evidence

Knowledge base

paper pages + entities + graph

Exploration tools

Agentic RAG · Multi-Graph · Co-author

Narrative: first show the pipeline, then explain the algorithm, then show how the digested database is consumed.

Start with the big picture: this is a developing AC system

The point is not only to classify papers once. The point is to maintain a living AC research knowledge system.

1

Broad paper pool

search pulls in thousands of candidate papers

2

Automated classification

papers are assigned relevance levels

3

AC knowledge base

AC-related papers enter the structured corpus

4

Evidence + Wiki

each paper gets reasons, scores, and a digest page

5

Graph + Q&A tools

Agentic RAG, Multi-Graph, Co-author, People Grid consume the corpus

paper → classification label + confidence + evidence → wiki/entity graph → retrieval/network/analytics

The system separates AC attribution from broader SDL relevance

A paper can be relevant to the field without being attributable to AC.

CORE_RELEVANT

high-confidence AC / SDL core

RELATED_RELEVANT

clear AC relation, weaker topic/core evidence

SDL_FIELD_RELEVANT

SDL field paper, weak AC attribution

ADJACENT

weak but potentially useful related signal

UNCERTAIN

needs reranker / judge / human review

EXCLUDE

no enough attribution or relevance signal

AC-positive labels are what move directly into the AC knowledge-base workflow; other labels support review, field monitoring, and exclusion.

The score is interpretable: evidence signals are combined, then rules decide the label

This is the part to explain after showing that the system can classify broad paper batches.

CLASSIFICATION CONFIDENCE

$$\text{confidence}(paper) = \Sigma\ w_i \cdot \text{signal}_i(paper)$$

signals = author match · institution match · topic relevance · network proximity · full-text evidence · citation evidence

AC attribution score

AC ≈ acknowledgement + funding + AC member + AC program + retrieval signal

SDL relevance score

SDL ≈ architecture signal + retrieval signal + SDL topic / keyword signal

Review gate

High evidence promotes CORE / RELATED. Weak or conflicting evidence goes to review.

Every classified paper has a detail page explaining why it was selected

This is what makes the classification reviewable and traceable.

Evidence drawer

label + confidence | score grid | human decision

- Author / institution / topic / full-text / network / citation scores
- Matched AC people, institutions, topics, and evidence snippets
- Graph signals: second-degree collaborator paths, AC core citations
- Reviewer can confirm or correct the label with notes

Why it matters

- The system can answer: why is this paper AC-related?
- Each label has auditability: score + reasons + source evidence
- Human-confirmed labels become locked review knowledge
- This creates the paper-level foundation for the later graph and Q&A layer

Each AC paper becomes a page in the knowledge-base layer

The current system has the first step: digesting papers into structured, reusable knowledge.



digested paper = markdown/wiki sections + extracted entities + traceable evidence pointers

Talk track: today the system builds paper-level pages and entity structure; the next layer is a richer graph-backed Q&A experience.

Once the corpus is digested, users can ask questions over it

This page is the Q&A/search entry point: an ask box retrieves text evidence, graph context, and SQL facts from the AC database.

Ask box examples

- Which AC papers use Bayesian optimization?
- What automation platforms appear in self-driving lab papers?
- Which papers mention funding acknowledgements?
- Which papers connect NLP, automation, and materials discovery?

Returned result

- Answer synthesized from retrieved source chunks
- Source papers and evidence snippets
- BM25 / cosine / graph score bars
- NL-to-SQL plan: intent, SQL, structured rows
- Graph path context: coauthor route or AC core citation
- Proposition hits: uses_method, funded_by, cites, topic

The new layer adds NL-to-SQL without removing the original retrieval stack

This is the LLM retrieval highlight: free-form questions can hit passages, graph paths, propositions, and database facts.

QUERY ROUTER

```
query → chunks + graph paths + propositions + SQL facts
```

The system can answer from passages, structured graph context, extracted claims, and exact database rows.

TEXT RANKING

```
RRF(d) = 1/(60 + rankBM25)
       + 1/(60 + rankcosine)
```

BM25 handles exact terms; cosine handles semantic match.

GRAPH BOOST

```
final = text_score × (1 + 0.3·Gnorm)
```

Graph score gently promotes papers central to AC relationships.

NL-TO-SQL

```
natural language → constrained intent → read-only SQL → structured rows
```

Adds exact counts, lists, top authors, institutions, and review queue rows without replacing retrieval.

The graph page lets users explore the entities extracted from papers

Each node is an entity; each edge is a relationship chosen by the user.

NODE SET

$$V = \text{authors} \cup \text{institutions} \cup \text{papers} \cup \text{topics} \cup \text{methods} \cup \text{funders}$$

Nodes are entities extracted from paper pages and wiki records.

EDGE SET

$$E = \text{authored} \cup \text{coauthored} \cup \text{cites} \cup \text{topic} \cup \text{method} \cup \text{funding}$$

Edges are selected relationships; users choose which graph to compose.

Co-author view

author nodes + coauthored edges

Citation view

paper / AC core nodes + cites_ac_core edges

Topic view

paper / topic nodes + belongs_to_topic edges

Method/funder view

paper / method / funder nodes + propositions

It helps AC ask relationship questions that a table cannot answer

The value is selective graph composition: choose which entities and relationships matter for the question.

Question types

- Which topics connect to which institutions?
- Which papers cite AC core work?
- Which methods appear across multiple AC-related papers?
- Which funders or programs are visible in evidence?

Interaction

- Toggle node types
- Toggle edge types
- Use TimeSlider by year
- Click a node to focus ego graph
- Group by institution, topic, country, era, label, community

Result

A visual map of how AC-attributed papers connect people, places, topics, methods, funding, and citation evidence.

The co-author page helps decide who AC should pay attention to

It moves from visualization to prioritization: which people are important under a chosen lens?

Use case

- For a given topic or AC-relevant corpus, identify external collaborators or bridge authors
- Switch between author network and institution heatmap
- Click a person to see papers, collaborators, and importance explanation

Six dimensions

- AC proximity
- Relationship strength
- Excess centrality
- Cross-SDL diversity
- Citation affinity
- Recency

The importance score is adjustable, but normalized

If AC disagrees with the default lens, the page can change the weights and recompute.

$$\text{importance}(\text{person}) = \Sigma\, w_i \cdot \text{dimensioni}(\text{person})$$

$$\Sigma\, w_i = 1$$

AC attribution
closer to AC and attributable work

Bridge builders
connect AC-adjacent communities

Institution hubs
durable institution-spanning collaboration

Emerging connectors
recent activity + early bridge potential

Topic diffusion
methods moving across SDL contexts

Talk track: this is not a fixed black-box ranking. The lens can be adjusted depending on whether AC cares about attribution, bridges, emerging people, or topic diffusion.

The people view turns AC members into a structured periodic table

This gives a simple entry point into the member-centric view of the knowledge base.

Periodic table of people

- Each AC member is an element
- Symbol = two letters from surname
- Atomic number = number of attributed papers
- Color = top research topic / SDL category
- Click a person to see papers and focus the network

Next layer

- Member-specific paper list
- Personal collaboration graph
- Topic / method / institution context
- Long-term view of how AC influence radiates through the global research network

The demo should end with decisions, not just features

These choices determine how the system should behave over repeated update cycles.

- Label policy: what exactly counts as AC, SDL-field, adjacent, uncertain, and exclude?
- Evidence standard: when can a paper automatically enter the AC knowledge base?
- Review policy: who reviews high-priority non-gold / uncertain papers, and how often?
- Graph policy: which relationships matter most for AC: coauthor, citation, topic, method, funding?
- People policy: which importance lens should guide outreach or collaboration analysis?
- Reporting policy: which metrics can be used in leadership or funder-facing claims?

Run the workflow on a real broad batch and calibrate with AC review

The current platform is ready to support the first operating-cycle discussion.

1

Run broad ingest

Use a 2,000-3,000 paper search result and let the system classify + build evidence pages.

2

Review border cases

Use evidence drawer, Wiki, Agentic RAG, and Multi-Graph to inspect uncertain or high-value papers.

3

Tune operating contract

Adjust thresholds, review rules, graph lenses, and people-importance profiles.

Target outcome: a defensible, repeatable AC literature attribution workflow that becomes smarter as the database grows.